Solution Requirements (Functional and Non-Functional)

Project: iMovies MERN **Document:** Solution Requirements (Functional and Non-Functional)

1. Purpose

This document lists functional requirements (what the system does) and non-functional requirements (quality and operational expectations) for the iMovies prototype.

2. Functional requirements

ID	Requirement	Priority
FR-01	The system shall allow a user to register with valid required details.	Must
FR-02	The system shall authenticate a registered user with valid credentials.	Must
FR-03	The system shall display movie records retrieved from the backend.	Must
FR-04	The system shall allow users to search movie listings.	Should
FR-05	The system shall allow a signed-in user to choose a movie, show date/time and seats.	Must
FR-06	The system shall calculate and display the booking total before submission.	Must
FR-07	The server shall validate booking requests and reject conflicting seat reservations when reservation controls are active.	Must
FR-08	The system shall allow a user to view only their own bookings.	Must
FR-09	The system shall allow cancellation only where permitted and only by the owner or an authorized admin.	Must
FR-10	The system shall protect admin operations with admin authentication/authorization.	Must
FR-11	An authorized admin shall be able to add and remove movie records.	Must
FR-12	The system shall provide clear success and error feedback.	Should
FR-13	The prototype shall clearly label any simulated payment as a demo, not a real financial transaction.	Must

3. Non-functional requirements

ID	Requirement	Verification idea
NFR-01 Usability	Labels and errors should be understandable to a first-time user.	Observe UAT participants.
NFR-02 Security	Passwords must be hashed; secrets kept in environment variables; protected endpoints enforce authorization.	Review code/configuration and test unauthorized requests.

ID	Requirement	Verification idea
NFR-03 Privacy	A user must not access another user’s booking records.	Attempt cross-user access in a controlled test.
NFR-04 Reliability	API should handle invalid requests without crashing.	Negative tests and server log review.
NFR-05 Performance	For a small local test dataset, common pages/API calls should feel responsive; set a measurable target before formal benchmarking.	Record timings under stated test conditions.
NFR-06 Maintainability	Frontend, backend and models should remain organized into understandable files/modules.	Code review.
NFR-07 Compatibility	UI should work in current desktop browsers and be checked at mobile widths.	Cross-browser and responsive checks.
NFR-08 Recoverability	Database backup/restore procedure should be defined before production use.	Restore test in non-production.
NFR-09 Deployment	Production configuration must use HTTPS, correct CORS, hosted database and secure secrets.	Deployment checklist.

4. Constraints

The prototype uses a MERN stack and depends on Node.js, npm, MongoDB and configured environment variables. Real payment processing and production hosting are not assumed to be available.

5. Acceptance mapping

FR-01 to FR-13 should be mapped to UAT cases. NFR-02 and NFR-03 require explicit security tests. Performance targets should be agreed before measuring; do not claim a performance result without recorded evidence.

6. Change management

Requirement ID: _____ Change requested: _____
Reason: _____ Priority: _____
Approved by: _____ Date: _____

Prepared for academic project documentation. Verify implementation-specific details and fill in actual test evidence before final submission.